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DSA0404 - FODS

1.

A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

Answer

Given:

- Age range = 18–75
- Income range = ₹20,000–₹20,00,000
- Age = 45
- Mean Age = 40
- SD = 12

Why failing to standardize distorts a distance-based algorithm

Distance-based algorithms such as K-Means and K-Nearest Neighbors (KNN) calculate distances between data points. If variables have very different scales, the variable with the larger numerical range contributes much more to the distance.

Here, Age ranges only from 18 to 75, while Income ranges from ₹20,000 to ₹20,00,000. Therefore, the numerical differences in Income are much larger than differences in Age.

For example, a difference of 10 years in Age may numerically be 10, while a difference of ₹1,00,000 in Income is 100,000. Without standardization, the distance calculation would therefore be dominated by Income, even if Age is equally important to the analysis.

Standardization converts variables to a common scale, generally with mean 0 and standard deviation 1. This allows each feature to contribute more fairly to the distance calculation.

Z-score calculation

$$Z = (X - \mu) / \sigma$$

Where:

$$X = 45$$

$$\mu = 40$$

$$\sigma = 12$$

$$Z = (45 - 40) / 12$$

$$Z = 5 / 12$$

$$Z = 0.417$$

Result: The z-score for Age = 45 is approximately 0.417, meaning the age is about 0.42 standard deviations above the mean.

2.

A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

Answer

Given:

- Mean = 45
- Median = 45
- Mode = 45
- Skewness ≈ 0
- Kurtosis = 5.8

Since the Mean, Median, and Mode are all equal to 45, the distribution is centered around 45. The skewness being approximately zero indicates that the distribution is approximately symmetric, with no substantial tendency toward either the left or right side.

However, these statistics alone do not describe how concentrated the observations are around the center or how heavy the tails are.

The important information comes from the kurtosis = 5.8.

Since the kurtosis is greater than 3, the distribution is leptokurtic. A leptokurtic distribution generally has:

- Heavier tails
- More extreme observations
- A sharper concentration around the center compared with a normal distribution

Therefore, even though the distribution is symmetric and has equal mean, median, and mode, it can still contain more extreme values than a normal distribution.

Key Insight:

Skewness tells us about asymmetry, while kurtosis tells us about tail heaviness/extreme observations. Thus, skewness ≈ 0 does not mean the distribution is necessarily normal.

Result: The distribution is approximately symmetric and centered at 45, but its kurtosis of 5.8 indicates a leptokurtic distribution with heavier tails and greater potential for extreme observations than a normal distribution.

3.

Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50

Frequency: 5, 8, 15, 12, 10

Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

Answer

Step 1: Find the midpoints

The midpoint is:

$$x_i = (\text{Lower} + \text{Upper}) / 2$$

Class Interval	Frequency (f)	Midpoint (x_i)
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Assumed mean: $A = 25$

Class width: $h = 10$

Step 2: Calculate u_i

$$u_i = (x_i - A) / h$$

x_i	f	u_i	fu_i	fu_i^2
5	5	-2	-10	20
15	8	-1	-8	8
25	15	0	0	0
35	12	1	12	12
45	10	2	20	40
Total	50		14	80

Step 3: Calculate Mean

Step-deviation formula:

$$\bar{x} = A + h(\Sigma fu / \Sigma f)$$

$$\bar{x} = 25 + 10(14 / 50)$$

$$= 25 + 2.8$$

$$= 27.8$$

Step 4: Calculate Variance

$$\sigma^2 = h^2 [\Sigma fu^2 / \Sigma f - (\Sigma fu / \Sigma f)^2]$$

$$\sigma^2 = 10^2 [80/50 - (14/50)^2]$$

$$= 100 [1.6 - (0.28)^2]$$

$$= 100 [1.6 - 0.0784]$$

$$= 100(1.5216)$$

$$= 152.16$$

Final Answer:

Mean = 27.8

Variance = 152.16

4.

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone doesn't.

Answer

Given:

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

Step 1: Coefficient of Variation

$$CV = (SD / \text{Mean}) \times 100$$

Portfolio A:

$$CV_a = (3 / 12) \times 100 \\ = 25\%$$

Portfolio B:

$$CV_\beta = (5 / 18) \times 100 \\ \approx 27.78\%$$

Portfolio	Mean Return	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Comparison

Portfolio B has a higher SD, so it is riskier in absolute terms.

However, when risk is measured relative to return using CV:

$$27.78\% > 25\%$$

Therefore, Portfolio B also has slightly higher relative risk per unit of return.

Evaluation of the claim

The first part of the claim is correct: Portfolio B is riskier in absolute terms because its SD is 5%, compared with 3% for Portfolio A.

However, the second part needs qualification. Portfolio B does provide a higher return of 18%, but its CV is also higher.

CV captures risk relative to the magnitude of the return, whereas SD only measures the absolute variability of returns.

Result: Portfolio B offers a higher return but also has a slightly higher relative risk. $CV_a = 25\%$ and $CV_\beta = 27.78\%$. Thus, based purely on CV, Portfolio A has the better risk-adjusted variability, while Portfolio B provides the higher expected return.

5.

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Answer

Given:

- Number of observations = 10
- Original variance = 25
- Original value = 40
- Approximate mean = 42
- Mistaken value = 400

(a) Effect on Mean

The mean will increase substantially because one observation has increased by 360.

Change in value:

$$400 - 40 = 360$$

Since there are 10 observations, the approximate change in the mean is:

$$360 / 10 = 36$$

So the new mean would be approximately:

$$42 + 36 = 78$$

This is a directional approximation because the original mean and variance are stated approximately.

Therefore, the mean will shift upward significantly.

(b) Effect on Variance

Variance will increase dramatically.

The reason is that variance is based on squared deviations from the mean:

$$\text{Variance} = \sum (x - \bar{x})^2 / N$$

Originally, the value 40 was close to the mean of 42:

$$40 - 42 = -2$$

Its squared deviation was only:

$$(-2)^2 = 4$$

After the recording error, the value becomes 400. Initially considering the old mean:

$$400 - 42 = 358$$

Its squared deviation becomes approximately:

$$358^2 = 128164$$

So the contribution of this single observation to the squared deviations changes from approximately 4 to 128,164 before accounting for the fact that the mean itself also shifts.

Why variance is affected much more than the mean

The mean uses the original values linearly:

$$\bar{x} = \sum x / N$$

Therefore, an extreme value affects the mean proportionally to its change.

Variance, however, uses squared deviations:

$$(x - \bar{x})^2$$

Therefore, a very large deviation becomes enormously larger after squaring.

Result:

- Mean: increases significantly because 40 is replaced by 400.
- Variance: increases disproportionately and dramatically because the extreme value produces a very large squared deviation.
- This demonstrates why variance and other squared-error measures are highly sensitive to outliers.

One-line takeaway: The mean is pulled toward the outlier, but variance is hit much harder because it squares the distance from the mean.